

Retail Sales Analytics: From SQL Exploration to Interactive Business Intelligence

1. Project Overview

This project analyzes a multi-year retail sales data warehouse using Microsoft SQL Server and Power BI. The objective is to move from raw dimensional tables to reproducible business analysis and then to an interactive decision-support dashboard. The analytical model is built around a sales fact table joined to customer and product dimensions, with a dedicated date dimension used in Power BI for time intelligence.

The workflow was designed as a connected analytics process rather than a collection of isolated queries. SQL was used first to understand the database structure, validate core measures, analyze magnitudes and rankings, and then extend the analysis with cumulative trends, year-over-year product performance, category contribution, and customer/product segmentation. Power BI reused the same business logic to create four report pages. Executive Overview, Sales Analysis, Product Analysis, and Customer Analysis.

Project Component	Implementation
Database / Querying	Microsoft SQL Server Management Studio
Analytical Model	gold.fact_sales, gold.dim_customers, gold.dim_products
Time Intelligence	Power BI dim_date table and DAX measures
Reporting	Interactive Power BI report with Year and Country slicers
Primary Analysis Themes	Sales trends, customer behavior, product performance, segmentation, ranking, and contribution analysis

2. Dataset and Analytical Model Summary

The project uses a star-schema style analytical model. The fact_sales table stores transactional measures such as sales amount, quantity, price, order number, order date, product key, and customer key. The customer and product dimensions provide descriptive attributes used for segmentation and reporting. Power BI adds a calendar table so yearly and monthly comparisons can be evaluated consistently.

Dataset Characteristic	Value/ Description
Total Sales	\$29.36M
Distinct Orders	Approximately 28K
Customers	18,484
Quantity Sold	Approximately 60K units
Products	295
Average Order Value	\$1.06K
Sales History	Multi-year history, approximately 2011 through early 2014 in the dashboard trend
Core Dimensions	Customer, product, and date

The use of dimension tables makes the model suitable for both descriptive exploration and higher-level business reporting. Customer-level measures can be analyzed by country, gender, age group, and behavioral segment, while product-level measures can be evaluated by category, cost range, performance group, and individual product.

3. Exploratory Data Analysis using SQL

The exploratory SQL stage establishes the structure and scale of the business before advanced calculations are introduced. The script begins with metadata exploration, then moves through dimensions, date coverage, business measures, magnitude analysis, and ranking analysis.

- **Database and Dimension Exploration:** INFORMATION_SCHEMA was used to inspect available tables and columns. Distinct country, category, subcategory, and product values were then reviewed to understand the descriptive dimensions available for reporting.
- **Date Exploration:** MIN(order_date), MAX(order_date), DATEDIFF() were used to understand the order history and the time span available for trend analysis.
- **Measures Exploration:** SUM(), AVG(), and COUNT(DISTINCT...) were used to establish total sales, total quantity, average price, total orders, total products, and total customers.
- **Magnitude Analysis:** Customer counts, products by category, average product cost, revenue by category, customer revenue, and quantity sold by country were compared to understand the relative size of business segments.
- **Ranking Analysis:** TOP, RANK(), AND ROW_NUMBER() were used to identify top products, bottom products, highest-value customers, and customers with fewest orders.

Business Question	SQL Technique	Dashboard Alignment
How large is the business?	SUM, AVG, COUNT(DISTINCT)	Executive KPI cards
Which categories generate the most revenue?	GROUP BY category + SUM(sales_amount)	Sales by Category
Where is the product volume concentrated?	GROUP BY country + SUM(quantity)	Quantity Sold by Country
Which products generate the most / least revenue?	TOP 5 and RANK()	Top 5 / Bottom 5 Products
Which customers generate the most revenue?	TOP 10 / RANK	Top 10 Customers by Sales
How are customers distributed by gender?	GROUP BY gender + COUNT	Customers by Gender

4. Advanced Analytics using SQL

The Advanced SQL stage moves beyond static totals and introduces time-series analysis, window functions, segmentation, and reusable reporting views. This is the part of the project that most directly supports the Sales, Product, and Customer Analysis pages in Power BI.

4.1 Changes over Time and Cumulative Analysis

Yearly and monthly revenue queries measure how sales change over time. A combined monthly query adds distinct customers and quantity, allowing revenue, customer activity, and sales volume to be tracked together. Window functions are then used to calculate cumulative sales and the moving average of product price.

```
SELECT
    DATETRUNC(month, order_date) AS [Order Date],
    SUM(sales_amount) AS [Revenue],
    COUNT(DISTINCT customer_key) AS [Total Customers],
    SUM(quantity) AS [Total Quantity]
FROM [dbo].[gold.fact_sales]
WHERE order_date IS NOT NULL
GROUP BY DATETRUNC(month, order_date)
ORDER BY [Order Date];
```

In the dashboard, this logic appears as Monthly Sales Trend, Running Total Sales, Average Price vs Moving Average, and Monthly Customer & Quantity Trend. The running total is cumulative across the historical period, so the 2013 table shows each month's current sales alongside lifetime cumulative sales to that point.

4.2 Product Performance and Year – over – Year Comparison

A yearly product-sales CTE combines GROUP BY with LAG() and windowed AVG() to compare each product against both its previous year and its own average performance. The output identifies whether sales increased or decreased and whether the current year is above or below the product's historical average.

2013 Sales Performance KPI	Value
Current Sales	\$16.34M
Previous Year Sales	\$5.84M
Sales Difference	\$10.50M
Year-over-Year Growth	179.8%

These measures are visualized on the Sales Analysis page and are also reused at the product level in the Product Performance Details table. This creates a direct bridge between SQL window-function analysis and interactive Power BI reporting.

4.3 Part-to-Whole Analysis and Segmentation

Category contribution is calculated by dividing each category's sales by overall sales. Product cost segmentation places products into Below 100, 100 – 500, 500 – 1000, and Above 1000 cost ranges. Customer segmentation uses order history and spending to classify customers as VIP, Regular, or New.

Customer Segment	Business Rule	Dashboard Count
VIP	Order history $\geq$ 12 months and total sales $>$ \$5,000	1,655
Regular	Order history $\geq$ 12 months and total sales $\leq$ \$5,000	2,198
New	Order history $<$ 12 months	14,631

The customer-report view extends this segmentation with recency, average order value, average monthly revenue, total orders, total quantity, and distinct products purchased. The product-report view performs a similar role for products by aggregating sales, orders, quantity, customers, lifespan, average selling price, and revenue-based performance categories.

5. SQL Refinements and Enhancements

The original SQL scripts provide the analytical foundation for the dashboard. During the report review, several places were identified where the logic can be made more accurate, rerunnable, or production-safe. The improvements below are recommendations, they are intentionally separated from the source queries rather than silently replacing them.

Area	Recommended Enhancement	Why It Improves the Analysis
Customers by Country	Replace SUM(customer_key) with COUNT(DISTINCT customer_key)	Customer keys are identifiers, so summing them does not measure the number of customers.
Monthly Performance	Group by DATETRUNC (month, order_date) instead of MONTH (order_date) alone	Prevents all Januarys, Februarys, etc. from different years being merged together.
Top 10 Customer Ranking	Use DENSE_RANK() or RANK() with a <= 10 filter, not <= 11	Keeps the ranking definition aligned with the business question.
Customer Segmentation	Use <= 5000 for Regular customers consistently in both the segmentation query and customer report.	Prevents exactly \$5,000 in sales from falling into the wrong category.
Age Category	Handle NULL birthdates as Unknown and use a fixed analysis date or dataset max date for historical reporting.	Improves reproducibility and avoids classifying missing ages as 50+.
Product Segmentation	Use High > 50,000; Mid > 10,000 and <= 50,000; Low <= 10,000	Corrects the original mid/low threshold inversion.
Product Report Counts	Use COUNT (DISTINCT customer_key) and COUNT(DISTINCT order_number)	Avoids counting line items as unique customers or unique orders.
Reusable Views	Use CREATE OR ALTER VIEW and NULLIF() in division expressions.	Makes scripts safe to return and reduces divide-by-zero risk.
Category Contribution	Order by the numeric contribution ratio before formatting it as text.	Avoids lexicographic sorting of percentage strings.

Recommended Product-Performance Logic:

```

CASE
  WHEN [Total Sales] > 50000 THEN 'High Performing Product'
  WHEN [Total Sales] > 10000 THEN 'Mid Performing Product'
  ELSE 'Low Performing Product'
END AS [Product Segmentation]

```

Recommended Customer-Report Logic:

CASE

WHEN [Age] IS NULL THEN 'Unknown'

WHEN [Age] < 20 THEN 'Under 20'

WHEN [Age] BETWEEN 20 AND 29 THEN '20-29'

WHEN [Age] BETWEEN 30 AND 39 THEN '30-39'

WHEN [Age] BETWEEN 40 AND 49 THEN '40-49'

ELSE '50 and Above'

END AS [Age Category],

CASE

WHEN [Order Timeframe] >= 12 AND [Total Sales] > 5000 THEN 'VIP Customers'

WHEN [Order Timeframe] >= 12 AND [Total Sales] <= 5000 THEN 'Regular Customers'

ELSE 'New Customers'

END AS [Customer Category]

6. Dashboard in Power BI

The Power BI report converts the SQL analysis into a business-facing interface. A consistent green visual theme is used across all four pages, while slicers allow the user to change the analytical context without rewriting SQL. Year filtering drives the Executive, Sales, and Product pages, while Customer page uses Country as its primary slicer.

6.1 Executive Overview

The Executive Overview summarizes business scale and high-level performance. KPI cards show \$29.36M in sales, approximately 28K orders, 18K customers, approximately 60K units sold, and \$1.06K average order value. The page then combines monthly sales, category revenue, country-level quantity and sales, gender distribution, country-level average order value, and a monthly customer-versus-quantity trend.

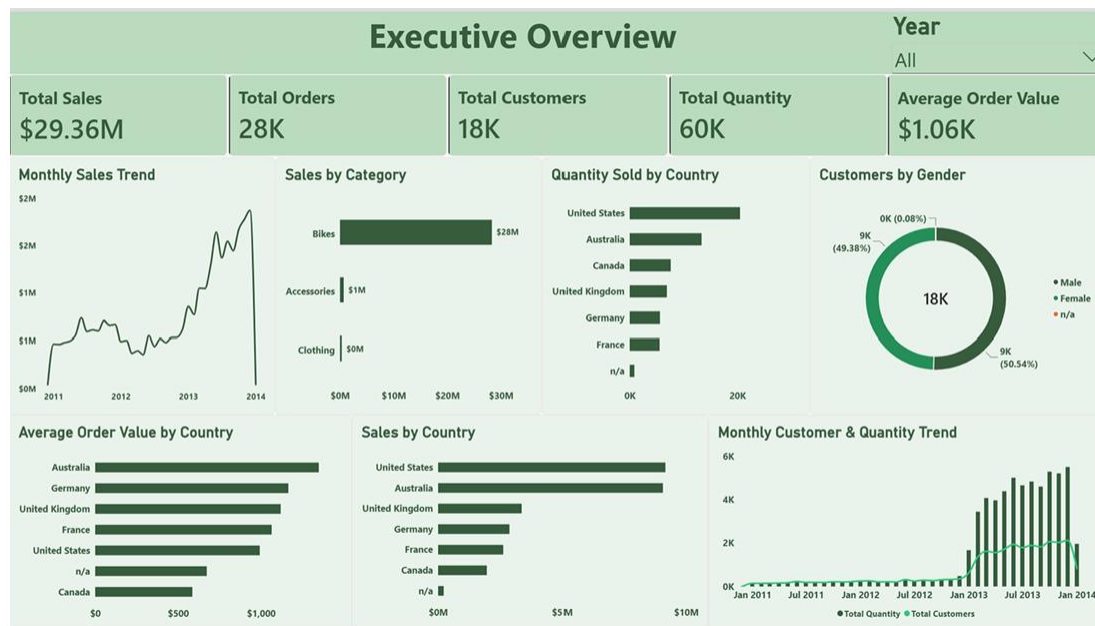


Figure 1. Executive Overview - overall business performance and geographic/customer summaries.

6.2 Sales Analysis

The Sales Analysis page focuses on time-series behavior and year-over-year performance. For 2013, current sales are \$16.34M compared with \$5.84M in the prior year, producing a \$10.50M increase and 179.8% year-over-year growth. The page also displays cumulative sales, average price versus moving average, a current-versus-previous-year comparison, and a detailed monthly performance table.

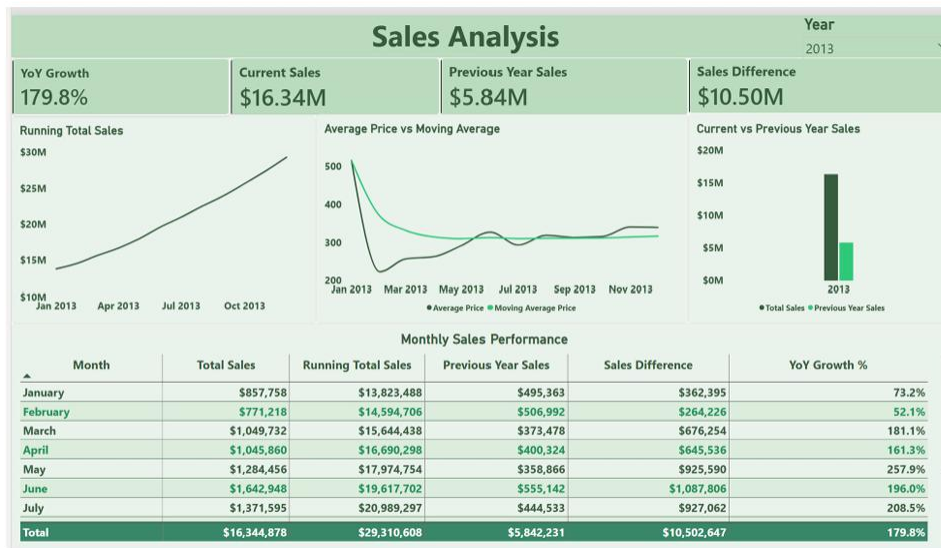


Figure 2. Sales Analysis - cumulative sales, moving average, YoY performance, and monthly detail.

The monthly table is particularly useful because it exposes the numbers behind the charts. It combines monthly revenue, cumulative sales, previous-year-sales, sales difference, and growth percentage in a single view, allowing the user to move from high-level visual interpretation to exact monthly values.

6.3 Product Analysis

The Product Analysis page combines the project's ranking, segmentation, contribution, and performance-analysis queries. In the 2013 view, Mountain-200 variants occupy the top sales positions, while several accessories and small items appear in the bottom five. The cost-range chart summarizes assortment structure, and the product-performance chart shows how products are distributed across High, Mid, and Low-performance groups.

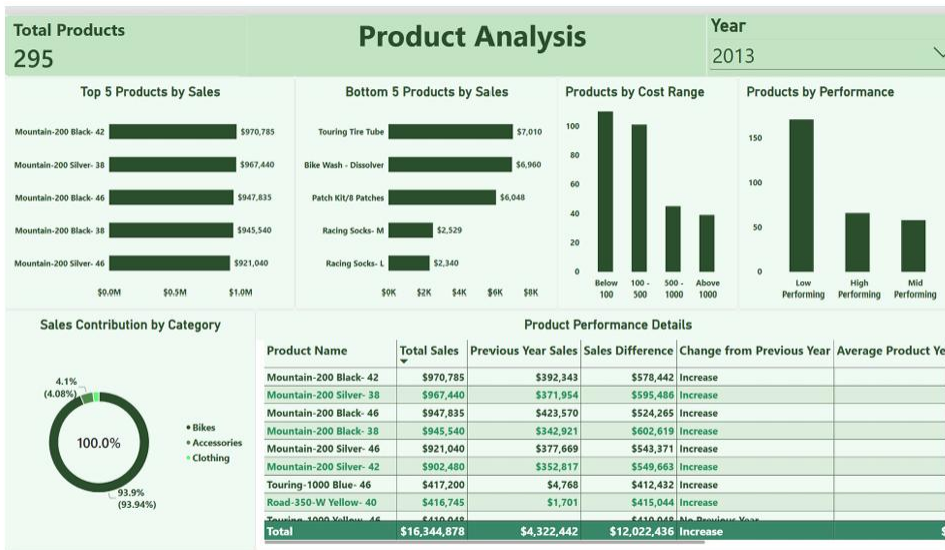


Figure 3. Product Analysis - product rankings, cost ranges, contribution, and performance details.

The Sales Contribution by Category visual shows a highly concentrated product mix, with Bikes contributing about 93.9% of 2013 sales. The Product Performance Details table then reproduces the advanced SQL logic at the product level: current sales, previous-year sales, difference, change direction, average yearly sales, difference from average, and performance versus average.

6.4 Customer Analysis

The Customer Analysis page operationalizes the customer-report view and the customer-ranking queries. The dataset contains 18,484 customers: 14,631 New, 2,198 Regular, and 1,655 VIP under the project's behavior-based segmentation rules. The page also shows age groups, segment counts, the top ten customers by sales, and a detailed customer table.

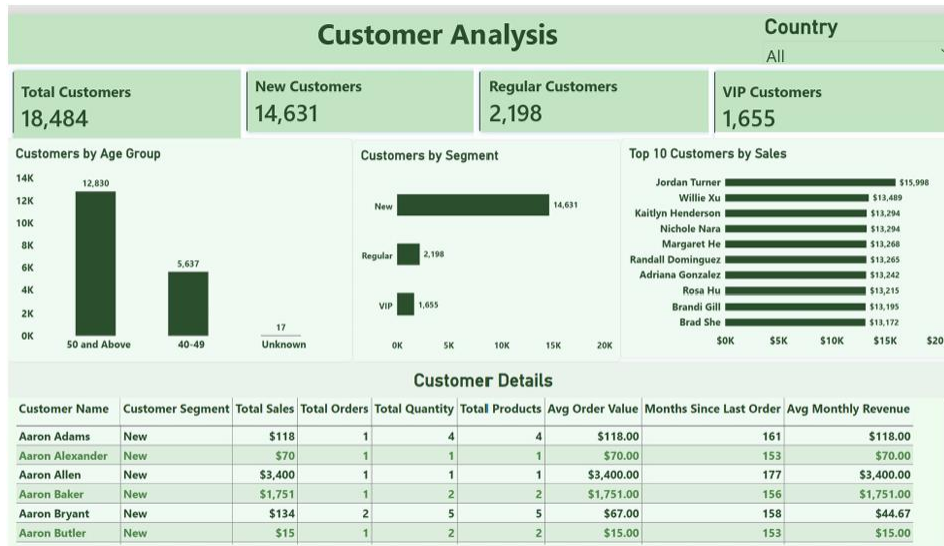


Figure 4. Customer Analysis - customer segmentation, age groups, top customers, and customer-level metrics.

Jordan Turner is the highest-value customer shown in the ranking at approximately \$15,998 in sales. The customer detail table extends the analysis with total sales, total orders, total quantity, total products, average order value, months since last order, and average monthly revenue. The Country slicer allows these measures to be evaluated by market.

Methodological note: age and recency are currently tied to the refresh date through TODAY() / GETDATE() – style logic. Because this is a historical sales dataset, these values will continue to increase as time passes. For a reproducible historical report, a fixe analysis date or MAX(order_date) should be used as the reference date.

7. Key Findings and Interpretations

- **Strong 2013 acceleration:** 2013 sales reached \$16.34M versus \$5.84M in the prior year, corresponding to 179.8% growth. The monthly charts also show a substantial increase in customer activity and quantity during 2013.
- **High category concentration:** Bikes contribute roughly 93.9% of 2013 sales, which explains why the highest-selling products are dominated by Mountain-200 bicycle variants.
- **Geographic concentration with different customer economics:** The United States and Australia lead total sales, while Australia has the highest average order value. This suggests volume leadership and value-per-order leadership are not identical.

- **Balanced gender distribution:** Customer gender is nearly evenly split between male and female records, so broad customer-level strategies should not assume a heavily skewed gender mix.
- **Large New-customer segment under the project definition:** 14,631 customers fall into the New segment because their observed order history is under 12 months. Only 1,655 customers qualify as VIP, leaving a substantial opportunity to extend customer lifespan and value.
- **Product performance is uneven:** The dashboard contains a meaningful group of high-performing products, but the low-performing group is the largest. The top/bottom rankings make the gap between strong and weak products highly visible.
- **Customer revenue is relatively dispersed at the top:** The leading customers generate roughly \$13K - \$16K each rather than one customer dominating the entire portfolio. This reduces single-customer concentration risk.

8. Business Recommendations

- **Protect the Bikes franchise while reducing concentration risk:** Bikes are the clear revenue engine. Inventory, availability, and promotional planning should prioritize this category, but management should also investigate whether Accessories and Clothing can be expanded to create a healthier revenue mix.
- **Use country-specific commercial strategies:** The United States is a volume and revenue leader, while Australia stands out in average order value. High-volume markets may respond to retention and repeat-purchase programs, whereas high – AOV markets may support premium bundles, upselling, or targeted product mixes.
- **Convert more New Customers into longer tenure segments:** Because the New segment is very large under the 12-month lifespan rule, lifecycle campaigns should encourage second purchases, longer customer tenure, and progression toward Regular and VIP status.
- **Use product rankings together with performance history:** Top-selling products should receive strong inventory support, while bottom products should be reviewed for low demand, product lifecycle stage, placement, or pricing. Year-over-year change and performance-versus-average should be considered before removing a product.
- **Improve analytical reproducibility:** Recency and age should be calculated relative to a controlled analysis date rather than the current system date when reporting on historical data. This prevents metrics from changing simply because the report is refreshed later.
- **Add margin and profitability data in the next iteration:** Current analysis focuses on revenue and volume. Cost-of-goods, gross margin, discount, and fulfillment cost data would allow management to distinguish high-revenue products from genuinely high-profit products.

9. Conclusion and Future Scope

This project demonstrates a complete SQL-to-BI workflow. Exploratory SQL establishes the database structure and core business measures; advanced SQL introduces time intelligence, window functions, segmentation, and reporting views; and Power BI converts those results into an interactive report that can be explored by year and country. The four report pages are closely aligned with the SQL questions rather than being separate presentation-only visuals.

The strongest technical value of the project is the connection between reusable SQL logic and interactive reporting. The Executive Overview summarizes the business, Sales Analysis explains how performance changes over time, Product Analysis connects ranking with contribution and historical performance, and Customer Analysis consolidates customer value, behavior, and recency metrics.

Future work can extend the project with profitability measures, cohorts, customer lifetime value, churn or repeat-purchase modeling, forecasting, product-margin optimization, and automated refresh. A production version could also move more of the reusable calculations into curated SQL views, use a controlled as-of-date for historical recency, and add row-level security or role-specific report views in Power BI.